Advanced Encryption Schemes

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0 Lecture 0

Definition 0.1 (KEM/DEM paradigm)**.**

$$\begin{aligned} \mathsf{KEM} &:= (\mathsf{Gen}, \mathsf{Encaps}, \mathsf{Decaps}), \\ \mathsf{Gen}(1^\lambda) &\longrightarrow (pk, sk), \\ \mathsf{Encaps}(pk) &\longrightarrow (K, c_1), \\ \mathsf{Decaps}(sk, c_1) &\longrightarrow K' \text{ or } \perp. \end{aligned} \qquad \begin{aligned} \mathsf{DEM} &:= (\mathsf{Enc}, \mathsf{Dec}), \\ \mathsf{Enc}(K, m) &\longrightarrow c_2, \\ \mathsf{Dec}(K, c_2) &\longrightarrow m' \text{ or } \perp. \end{aligned}$$

Hybrid encryption : $(K, c_1) \leftarrow \mathsf{KEM.Encaps}(pk), \quad c_2 \leftarrow \mathsf{DEM.Enc}(K, m), \quad ct := (c_1, c_2),$

Hybrid decryption : $K' \leftarrow \mathsf{KEM.Decaps}(sk, c_1), \quad m' \leftarrow \mathsf{DEM.Dec}(K', c_2) \quad (K' \neq \perp).$

$K' = \perp \implies$ hybrid decryption outputs $\perp$.

KEM : Key Encapsulation Mechanism,

DEM : Data Encapsulation Mechanism,

$\lambda \in \mathbb{N}$: security parameter,

pk, sk : public and secret keys,

K : symmetric key,

m : message,

c_1, c_2 : key encapsulation, encrypted data,

$\perp$: failure.

1 Lecture 1

Definition 1.1 (Probabilistic polynomial-time algorithm)**.**

$$\mathrm{PPT}(A) \iff \exists p \in \mathrm{poly} \quad \forall x \in \{0,1\}^* \quad \forall r : \ \mathrm{time}_A(x;r) \leq p(|x|).$$

Definition 1.2 (Negligible function)**.**

$$\mu : \mathbb{N} \to \mathbb{R} \text{ negligible} \iff \forall c \in \mathbb{N} \ \exists \lambda_0 \in \mathbb{N} \ \forall \lambda \geq \lambda_0 : \quad |\mu(\lambda)| < \frac{1}{\lambda^c}.$$

Exercise 1.1 (Exercise sheet 1: Negligible Functions)**.** Prove or disprove that the following functions are negligible. You can use that $f(n) = 1/2^n$ is negligible.

0. $f_0(\lambda) = \dfrac{1}{2^\lambda}$.

   Solution.

$$\begin{aligned} & \forall c \in \mathbb{N} : \quad \lim_{\lambda \to \infty} \frac{\lambda^c}{2^\lambda} = 0 \\ \implies & \forall c \in \mathbb{N} \ \exists \lambda_0 \in \mathbb{N} \ \forall \lambda \geq \lambda_0 : \quad \frac{\lambda^c}{2^\lambda} < 1 \\ \implies & \forall c \in \mathbb{N} \ \exists \lambda_0 \in \mathbb{N} \ \forall \lambda \geq \lambda_0 : \quad |f_0(\lambda)| = \frac{1}{2^\lambda} < \frac{1}{\lambda^c} \\ \implies & f_0 \text{ is negligible.} \end{aligned}$$

   □

1. $f_1(n) = \dfrac{1}{n^{20}}$.

   Solution.

$$\begin{aligned} c := 20 \in \mathbb{N} \implies & \forall \lambda_0 \in \mathbb{N} \ \exists \lambda \geq \lambda_0 : \quad |f_1(\lambda)| = \frac{1}{\lambda^{20}} = \frac{1}{\lambda^c} \geq \frac{1}{\lambda^c} \\ \implies & \exists c \in \mathbb{N} \ \forall \lambda_0 \in \mathbb{N} \ \exists \lambda \geq \lambda_0 : \quad |f_1(\lambda)| \geq \frac{1}{\lambda^c} \\ \implies & f_1 \text{ is non-negligible.} \end{aligned}$$

   □

2. $f_2(n) = \dfrac{2^n}{(2^n)^2}$.

   Solution.

$$|f_2(\lambda)| = \frac{2^\lambda}{(2^\lambda)^2} = \frac{1}{2^\lambda} = f_0(\lambda) \implies f_2 \text{ is negligible.}$$

   □

3. $f_3(n) = \left|\dfrac{(-1)^n}{2^n}\right|$.

   Solution.

$$|f_3(\lambda)| = \left|\frac{(-1)^\lambda}{2^\lambda}\right| = \frac{1}{2^\lambda} = f_0(\lambda) \implies f_3 \text{ is negligible.}$$

   □

4. $f_4(n) = \dfrac{(-1)^n}{2^n}$.

Solution.

$$|f_4(\lambda)| = \left|\frac{(-1)^\lambda}{2^\lambda}\right| = \frac{1}{2^\lambda} = f_0(\lambda) \implies f_4 \text{ is negligible.}$$

□

5. $f_5(n) = \dfrac{n^2 + n^7}{2^n}$.

Solution.

$$\begin{aligned}
& \forall c \in \mathbb{N}: \quad \lim_{\lambda\to\infty} \lambda^c |f_5(\lambda)| = \lim_{\lambda\to\infty} \left(\frac{\lambda^{c+2}}{2^\lambda} + \frac{\lambda^{c+7}}{2^\lambda}\right) = 0 \\
\implies & \forall c \in \mathbb{N}\ \exists \lambda_0 \in \mathbb{N}\ \forall \lambda \geq \lambda_0: \quad \lambda^c |f_5(\lambda)| < 1 \\
\implies & \forall c \in \mathbb{N}\ \exists \lambda_0 \in \mathbb{N}\ \forall \lambda \geq \lambda_0: \quad |f_5(\lambda)| < \frac{1}{\lambda^c} \\
\implies & f_5 \text{ is negligible.}
\end{aligned}$$

□

Definition 1.3 (Public key encryption scheme)**.**

$$\Pi := (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec}).$$

$$\mathrm{PKE}(\Pi) \iff \begin{cases}
\mathrm{PPT}(\mathsf{Gen}) \wedge \mathrm{PPT}(\mathsf{Enc}) \wedge \mathrm{PPT}(\mathsf{Dec}), \\
\forall \lambda \in \mathbb{N}: \\
\quad \mathcal{K}_{\mathrm{pub},\lambda} \times \mathcal{K}_{\mathrm{sec},\lambda} \ni (pk, sk) \longleftarrow \mathsf{Gen}(1^\lambda), \\
\quad \forall pk \in \mathcal{K}_{\mathrm{pub},\lambda}\ \forall m \in \mathcal{M}_\lambda: \quad \mathcal{C}_\lambda \ni ct \longleftarrow \mathsf{Enc}(pk, m), \\
\quad\quad \forall sk \in \mathcal{K}_{\mathrm{sec},\lambda}\ \forall ct \in \mathcal{C}_\lambda: \quad \mathcal{M}_\lambda \cup \{\perp\} \ni m' \longleftarrow \mathsf{Dec}(sk, ct).
\end{cases}$$

$\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec}$: key generation, encryption, decryption algorithms,
λ : security parameter,
$\mathcal{K}_{\mathrm{pub},\lambda}, \mathcal{K}_{\mathrm{sec},\lambda}$: public and secret key spaces,
$\mathcal{M}_\lambda, \mathcal{C}_\lambda$: message and ciphertext spaces,
pk, sk : public and secret keys,
m, ct, m' : message, ciphertext, decryption output,
$\perp$: decryption failure.

$$\mathcal{K}_{\mathrm{pub},\lambda}, \mathcal{K}_{\mathrm{sec},\lambda}, \mathcal{M}_\lambda, \mathcal{C}_\lambda \subseteq \{0,1\}^*, \qquad \perp \notin \{0,1\}^*.$$

Definition 1.4 (Perfect correctness)**.**

$$\mathrm{PerfectCorrect}(\Pi) \iff \forall \lambda \in \mathbb{N}\ \forall (pk, sk) \in \mathsf{Gen}(1^\lambda)\ \forall m \in \mathcal{M}_\lambda:$$

$$\Pr[\mathsf{Dec}(sk, \mathsf{Enc}(pk, m)) = m] = 1.$$

Definition 1.5 (Perfect security)**.**

$$\begin{aligned}
\mathrm{PerfectSecure}(\Pi) \iff & \forall \lambda \in \mathbb{N}\ \forall (pk, sk) \in \mathsf{Gen}(1^\lambda) \\
& \forall m_0, m_1 \in \mathcal{M}_\lambda\ \forall ct \in \mathcal{C}_\lambda: \\
& |m_0| = |m_1| \implies \Pr_r[\mathsf{Enc}(pk, m_0; r) = ct] = \Pr_r[\mathsf{Enc}(pk, m_1; r) = ct].
\end{aligned}$$

Π : public key encryption scheme,
λ : security parameter,
pk, sk : generated public and secret keys,
$\mathcal{M}_\lambda, \mathcal{C}_\lambda$: message and ciphertext spaces,
m_0, m_1 : messages of equal length,
ct : ciphertext,
$|m|$: bit length of m,
r : encryption's random bits,
$\Pr_r$: probability over r.

Definition 1.6 (Perfect security: conditional probabilities)**.**

$$\mathcal{M}_{\lambda,\ell} := \{m \in \mathcal{M}_\lambda : |m| = \ell\}.$$

$$\begin{aligned}\text{PerfectSecure}_{\text{cond}}(\Pi) \iff\ & \forall \lambda \in \mathbb{N}\ \forall (pk, sk) \in \mathsf{Gen}(1^\lambda)\ \forall \ell \in \mathbb{N} \cup \{0\} \\ & \forall D \text{ on } \mathcal{M}_{\lambda,\ell}\ \forall m \in \mathcal{M}_{\lambda,\ell}\ \forall ct \in \mathcal{C}_\lambda : \\ & \Pr[PK = pk, C = ct] > 0 \implies \\ & \quad \Pr[M = m \mid PK = pk, C = ct] = \Pr[M = m], \\ & (PK, SK) \leftarrow \mathsf{Gen}(1^\lambda), \quad M \leftarrow D, \quad C \leftarrow \mathsf{Enc}(PK, M).\end{aligned}$$

ℓ : fixed message length,
$\mathcal{M}_{\lambda,\ell}$: messages of length ℓ,
D : arbitrary message distribution,
PK, SK : public and secret key random variables,
pk : public key value on which we condition,
M, C : message and ciphertext random variables,
Pr : probability over independent key generation, message sampling, and encryption randomness.

Lemma 1.7 (Equivalence of perfect-security definitions)**.**

$$\text{PerfectSecure}(\Pi) \iff \text{PerfectSecure}_{\text{cond}}(\Pi).$$

Proof.

$\lambda,\ (pk, sk),\ \ell$: fixed allowed parameter, generated keys, and length,

$$\Pr_{pk}[E] := \Pr[E \mid PK = pk],$$
$$q_m(ct) := \Pr_r[\mathsf{Enc}(pk, m; r) = ct],$$
$$p_m := \Pr[M = m] = \Pr_{pk}[M = m] \qquad (M \perp PK).$$

$$\begin{aligned}(\implies) \quad & q_m(ct) = q(ct) \quad \forall m \in \mathcal{M}_{\lambda,\ell} \\ \implies \quad & \Pr_{pk}[C = ct] = \sum_{u \in \mathcal{M}_{\lambda,\ell}} p_u q_u(ct) = q(ct) \\ \implies \quad & \Pr_{pk}[M = m \mid C = ct] = \frac{p_m q_m(ct)}{\Pr_{pk}[C = ct]} = p_m \quad (\Pr_{pk}[C = ct] > 0).\end{aligned}$$

$$
\begin{aligned}
(\Longleftarrow)\quad & m_0 \neq m_1, \quad |m_0| = |m_1| = \ell, \quad D := \mathrm{Uniform}(\{m_0, m_1\}), \\
& s := q_{m_0}(ct) + q_{m_1}(ct), \qquad \Pr_{pk}[C = ct] = s/2, \\
& s > 0 \implies \frac{q_{m_0}(ct)}{s} = \Pr_{pk}[M = m_0 \mid C = ct] = \frac{1}{2} \implies q_{m_0}(ct) = q_{m_1}(ct), \\
& s = 0 \implies q_{m_0}(ct) = q_{m_1}(ct) = 0.
\end{aligned}
$$

□

Lemma 1.8 (Key-space bound for PKE)**.**

$$\mathcal{K}_{\mathrm{sec},\lambda}(pk) := \{sk : (pk, sk) \in \mathsf{Gen}(1^\lambda)\}.$$

$$
\begin{aligned}
& \mathrm{PerfectCorrect}(\Pi) \wedge \mathrm{PerfectSecure}_{\mathrm{cond}}(\Pi) \\
\implies \quad & \forall \lambda \in \mathbb{N}\ \forall (pk, sk) \in \mathsf{Gen}(1^\lambda)\ \forall \ell \in \mathbb{N} \cup \{0\} : \quad |\mathcal{K}_{\mathrm{sec},\lambda}(pk)| \geq |\mathcal{M}_{\lambda,\ell}|.
\end{aligned}
$$

$\mathcal{K}_{\mathrm{sec},\lambda}(pk)$: *generated secret keys compatible with* pk,
$\mathcal{M}_{\lambda,\ell}$: *supported messages of length* ℓ.

Proof.

$$\lambda, pk, \ell : \text{ fixed}, \quad \mathcal{M}_{\lambda,\ell} \neq \varnothing, \quad M \leftarrow \mathrm{Uniform}(\mathcal{M}_{\lambda,\ell}), \qquad ct : \ \Pr[PK = pk, C = ct] > 0.$$

$$
\begin{aligned}
\forall m \in \mathcal{M}_{\lambda,\ell} : \quad & \Pr[M = m \mid PK = pk, C = ct] = \frac{1}{|\mathcal{M}_{\lambda,\ell}|} > 0 \\
\implies \quad & \exists sk_m \in \mathcal{K}_{\mathrm{sec},\lambda}(pk) : \quad \Pr[PK = pk, SK = sk_m, M = m, C = ct] > 0 \\
\implies \quad & \Pr_s[\mathsf{Dec}(sk_m, ct; s) = m] = 1.
\end{aligned}
$$

$$
\begin{aligned}
m \neq m' &\implies sk_m \neq sk_{m'} \\
&\implies (m \mapsto sk_m) \text{ is injective} \\
&\implies |\mathcal{K}_{\mathrm{sec},\lambda}(pk)| \geq |\mathcal{M}_{\lambda,\ell}|. \\
\mathcal{M}_{\lambda,\ell} = \varnothing &\implies |\mathcal{K}_{\mathrm{sec},\lambda}(pk)| \geq 0.
\end{aligned}
$$

□

$$
\begin{aligned}
\mathcal{M}_\lambda = \{0,1\}^*, \quad |sk| \leq p(\lambda) \quad (p \in \mathrm{poly}) &\implies |\mathcal{K}_{\mathrm{sec},\lambda}(pk)| \leq 2^{p(\lambda)+1} - 1, \\
\ell > p(\lambda) + 1 &\implies |\mathcal{M}_{\lambda,\ell}| = 2^\ell > |\mathcal{K}_{\mathrm{sec},\lambda}(pk)|.
\end{aligned}
$$

Exercise 1.3 (Exercise sheet 1: Perfect Secrecy)**.** Prove that perfect secrecy is impossible for PKE schemes.

$$
\begin{aligned}
& \mathrm{PerfectCorrect}(\Pi), \quad \lambda \in \mathbb{N}, \quad (pk, sk) \in \mathsf{Gen}(1^\lambda), \\
& m_0, m_1 \in \mathcal{M}_\lambda, \quad m_0 \neq m_1, \quad |m_0| = |m_1| = \ell.
\end{aligned}
$$

Solution 1: ciphertext distributions.

$$q_i(ct) := \Pr_r[\mathsf{Enc}(pk, m_i; r) = ct] \qquad (i \in \{0,1\}).$$

$$
\begin{aligned}
&\exists ct \in \mathcal{C}_\lambda : \\
&\quad q_0(ct) > 0 \implies \Pr_s[\mathsf{Dec}(sk, ct; s) = m_0] = 1, \\
&\quad q_1(ct) > 0 \implies \Pr_s[\mathsf{Dec}(sk, ct; s) = m_1] = 1, \\
&\quad m_0 \neq m_1 \implies \Pr_s[\mathsf{Dec}(sk, ct; s) = m_0] + \Pr_s[\mathsf{Dec}(sk, ct; s) = m_1] \leq 1 \\
&\quad \implies \big(q_1(ct) = 0 \wedge q_0(ct) > 0\big) \vee \big(q_1(ct) > 0 \wedge q_0(ct) = 0\big) \implies q_1(ct) \neq q_0(ct) \\
&\quad \implies \neg\text{PerfectSecure}(\Pi).
\end{aligned}
$$

r, s : encryption and decryption random bits.

□

Solution 2: conditional probabilities.

$$
\begin{aligned}
\mathsf{Find}_\lambda(pk) : \quad & r \in \{0,1\}^* \text{ enumerated by increasing length :} \\
& \mathsf{Gen}(1^\lambda; r) = (pk, sk') \implies \text{return } sk', \\
A_\lambda(pk, ct) &:= \mathsf{Dec}(\mathsf{Find}_\lambda(pk), ct).
\end{aligned}
$$

$$
\begin{aligned}
(pk, sk) \in \mathsf{Gen}(1^\lambda) &\implies \mathsf{Find}_\lambda(pk) \text{ terminates}, \\
&\implies (pk, \mathsf{Find}_\lambda(pk)) \in \mathsf{Gen}(1^\lambda), \\
&\implies \Pr[A_\lambda(pk, \mathsf{Enc}(pk, m)) = m] = 1.
\end{aligned}
$$

$$
\begin{aligned}
&D := \text{Uniform}(\{m_0, m_1\}), \quad (PK, SK) \leftarrow \mathsf{Gen}(1^\lambda), \quad M \leftarrow D, \quad C \leftarrow \mathsf{Enc}(PK, M), \\
&\Pr[A_\lambda(PK, C) = M] = 1, \qquad \Pr[M = m_0] = \tfrac{1}{2}, \\
&ct : \quad \Pr_r[\mathsf{Enc}(pk, m_0; r) = ct] > 0 \\
&\implies \quad \Pr[PK = pk, C = ct] > 0, \\
&\Pr[A_\lambda(pk, ct) = m_0] = 1 \\
&\implies \quad \Pr[M = m_0 \mid PK = pk, C = ct] = 1 \neq \tfrac{1}{2} = \Pr[M = m_0] \\
&\implies \quad \neg\text{PerfectSecure}_{\text{cond}}(\Pi).
\end{aligned}
$$

Find_λ : exhaustive search for a compatible secret key,
A_λ : attacker with no polynomial runtime restriction,
D : message distribution independent of key generation.

□

Definition 1.9 (Semantic security)**.**

$$
\begin{aligned}
\text{SemanticSecure}(\Pi) \iff & \forall \text{ PPT } A \; \exists \text{ PPT } A' \; \forall \text{ PPT } \mathsf{Samp} \; \forall \text{ PT } f : \\
& \forall \lambda \in \mathbb{N} : \quad |\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| \leq \text{negl}(\lambda).
\end{aligned}
$$

$$
\epsilon_A(\lambda) := \Pr\left[\begin{array}{ll} (pk, sk) \leftarrow \mathsf{Gen}(1^\lambda) & : \; A(pk, h, \mathsf{Enc}(pk, m)) = f(m, h) \\ (m, h) \leftarrow \mathsf{Samp}(pk) & \end{array} \right],
$$

$$
\epsilon_{A'}(\lambda) := \Pr\left[\begin{array}{ll} (pk, sk) \leftarrow \mathsf{Gen}(1^\lambda) & : \; A'(pk, h, |m|) = f(m, h) \\ (m, h) \leftarrow \mathsf{Samp}(pk) & \end{array} \right].
$$

Π : public key encryption scheme,
A : attacker given the public key, hint, and ciphertext,
A' : simulator given the public key, hint, and message length,
Samp : sampler returning (m, h) with $m \in \mathcal{M}_\lambda$,
h : hint containing public information about m,
f : deterministic polynomial-time function of (m, h),
$\epsilon_A, \epsilon_{A'}$: probabilities of correctly computing $f(m, h)$,
negl : negligible bound,
λ : security parameter,
Pr : probability over all random bits in the experiment.

Definition 1.10 (Security game)**.**

$$\mathsf{G}^A(1^\lambda) \longrightarrow w \in \{0,1\}, \qquad A \text{ wins} \iff w = 1.$$

G : randomized experiment specifying a challenger, permitted interactions, and a winning condition,
A : attacker interacting with the challenger,
λ : security parameter,
w : experiment's output bit; 1 means a win,
$\Pr[\mathsf{G}^A(1^\lambda) = 1]$: winning probability over all participants' random bits.

Definition 1.11 (IND-CPA game)**.**

$\mathsf{G}^{\text{IND-CPA}}_{\Pi,A}(1^\lambda)$:

Challenger		Attacker $A = (A_0, A_1)$
$(pk, sk) \leftarrow \mathsf{Gen}(1^\lambda)$	$\xrightarrow{pk}$	
		$(m_0, m_1, st) \leftarrow A_0(pk)$
		$m_0 \neq m_1 \in \mathcal{M}_\lambda, \quad \lvert m_0\rvert = \lvert m_1\rvert$
	$\xleftarrow{m_0, m_1}$	
$b \leftarrow_\$ \{0,1\}$		
$ct \leftarrow \mathsf{Enc}(pk, m_b)$	$\xrightarrow{ct}$	
		$b^* \leftarrow A_1(st, ct)$
	$\xleftarrow{b^*}$	
return $\mathbf{1}[b^* = b]$		

Π : public key encryption scheme,
λ : security parameter,
$\mathsf{G}^{\text{IND-CPA}}_{\Pi,A}$: indistinguishability game under chosen-plaintext attack,
$A = (A_0, A_1)$: two-stage attacker,
A_0 : attacker selecting $m_0 \neq m_1 \in \mathcal{M}_\lambda,\ |m_0| = |m_1|$,
A_1 : attacker returning a guess $b^* \in \{0,1\}$,
admissible A : attacker always satisfying these message and guess conditions,
st : attacker's state passed from A_0 to A_1,
b : uniformly sampled hidden bit,
ct : challenge ciphertext encrypting m_b,
$\leftarrow_\$$: uniform sampling,
$\mathbf{1}[E]$: 1 if E holds, and 0 otherwise.

Definition 1.12 (IND-CPA security)**.**

$$P^{\text{IND-CPA}}_{\Pi,A,\text{win}}(\lambda) := \Pr\left[\mathsf{G}^{\text{IND-CPA}}_{\Pi,A}(1^\lambda) = 1\right] = \Pr[b^* = b],$$
$$\text{Adv}^{\text{IND-CPA}}_{\Pi,A}(\lambda) := 2\left|P^{\text{IND-CPA}}_{\Pi,A,\text{win}}(\lambda) - \frac{1}{2}\right|.$$

$$\text{IND-CPA-Secure}(\Pi) \iff \forall \text{ admissible PPT } A = (A_0, A_1) : \quad \forall \lambda \in \mathbb{N} : \quad \text{Adv}^{\text{IND-CPA}}_{\Pi,A}(\lambda) \leq \text{negl}(\lambda).$$

Π : public key encryption scheme,
IND : indistinguishability,
CPA : chosen-plaintext attack,
A : admissible attacker in the IND-CPA game,
$P^{\text{IND-CPA}}_{\Pi,A,\text{win}}(\lambda)$: winning probability in the IND-CPA game,
$\text{Adv}^{\text{IND-CPA}}_{\Pi,A}$: twice the absolute deviation from random-guess success,
negl : negligible bound,
Pr : probability over key generation, encryption, b, and A's randomness.

Definition 1.13 (Concrete IND-CPA security; lecture Definition 1.5)**.**

$$\Pi \text{ is } (t,\epsilon)\text{-IND-CPA secure} \iff \forall \text{ admissible PPT } A \; \forall \lambda \in \mathbb{N} : \quad T_A(\lambda) \leq t(\lambda) \implies \text{Adv}^{\text{IND-CPA}}_{\Pi,A}(\lambda) \leq \epsilon(\lambda).$$

Π : public key encryption scheme,
A : admissible attacker in the IND-CPA game,
λ : security parameter,
$T_A(\lambda)$: worst-case total runtime of A_0 and A_1 in the game,
$t(\lambda)$: attacker runtime bound,
$\epsilon(\lambda)$: advantage bound,
$\text{Adv}^{\text{IND-CPA}}_{\Pi,A}(\lambda)$: advantage defined above.

Theorem 1.14 (Semantic security $\iff$ IND-CPA)**.**

$$\text{SemanticSecure}(\Pi) \iff \text{IND-CPA-Secure}(\Pi).$$

Π : *public key encryption scheme.*

First direction: semantic security $\implies$ IND-CPA.

$$\text{Contrapositive goal:} \quad \left[\exists \text{ PPT } D : \quad \text{Adv}^{\text{IND-CPA}}_{\Pi,D} \text{ non-negligible}\right] \implies \begin{bmatrix}\exists \text{ PPT } A \; \forall \text{ PPT } A' \; \exists \text{ PPT } \mathsf{Samp} \; \exists \text{ PT } f : \\ |\epsilon_A - \epsilon_{A'}| \text{ non-negligible}\end{bmatrix}.$$

$$\text{Idea:} \quad \begin{bmatrix}\exists \text{ PPT } A \; \exists \text{ PPT } \mathsf{Samp} \; \exists \text{ PT } f \; \forall \text{ PPT } A' : \\ \exists \text{ polynomial } p \geq 1 \; \forall \lambda \in \mathbb{N} : \\ |\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| \geq \dfrac{\text{Adv}^{\text{IND-CPA}}_{\Pi,D}(\lambda)}{p(\lambda)} - \text{negl}(\lambda)\end{bmatrix}$$
$$\overset{\text{Adv}^{\text{IND-CPA}}_{\Pi,D} \text{ non-negligible}}{\implies} \quad \forall \text{ PPT } A' : \quad |\epsilon_A - \epsilon_{A'}| \text{ non-negligible.}$$

$$\neg\text{IND-CPA-Secure}(\Pi) \implies \exists \text{ PPT } D = (D_0, D_1) : \quad \mathrm{Adv}^{\text{IND-CPA}}_{\Pi,D} \text{ non-negligible.}$$

$$\begin{aligned}
&D^{(0)} := (D_0, D_1), \qquad D^{(1)} := (D_0, 1 - D_1),\\
&P^{\text{IND-CPA}}_{\Pi,D^{(1)},\text{win}}(\lambda) = 1 - P^{\text{IND-CPA}}_{\Pi,D^{(0)},\text{win}}(\lambda),\\
&\max_{i\in\{0,1\}} P^{\text{IND-CPA}}_{\Pi,D^{(i)},\text{win}}(\lambda) = \frac{1}{2} + \frac{1}{2}\mathrm{Adv}^{\text{IND-CPA}}_{\Pi,D}(\lambda),\\
&\implies \quad \exists i \in \{0,1\} :\\
&\qquad P^{\text{IND-CPA}}_{\Pi,D^{(i)},\text{win}} - \frac{1}{2} \text{ non-negligibly positive},\\
&D := D^{(i)}.
\end{aligned} \tag{1}$$

$$\begin{aligned}
\mathsf{Samp}(pk): \quad & (m_0, m_1, st) \leftarrow D_0(pk);\\
& b \leftarrow_\$ \{0,1\};\\
& \text{return } (m_b, (m_0, m_1, st));
\end{aligned}$$

$$f(m_b, (m_0, m_1, st)) := \mathbf{1}[m_b \neq m_0] = b;$$

$$A(pk, (m_0, m_1, st), ct) := D_1(st, ct).$$

$$\epsilon_A(\lambda) = \Pr\left[\begin{array}{l}(pk, sk) \leftarrow \mathsf{Gen}(1^\lambda)\\ (m_b, h) \leftarrow \mathsf{Samp}(pk)\end{array} \;:\; A(pk, h, \mathsf{Enc}(pk, m_b)) = f(m_b, h)\right]$$

$$= \Pr\left[\begin{array}{l}(pk, sk) \leftarrow \mathsf{Gen}(1^\lambda)\\ (m_0, m_1, st) \leftarrow D_0(pk)\\ b \leftarrow_\$ \{0,1\}\end{array} \;:\; D_1(st, \mathsf{Enc}(pk, m_b)) = b\right]$$

$$= P^{\text{IND-CPA}}_{\Pi,D,\text{win}}(\lambda).$$

$$\begin{aligned}
&\forall \text{ PPT } A' : \quad b \text{ independent of } (pk, h, |m_b|),\\
&\qquad Y := A'(pk, h, |m_b|),\\
&\qquad \epsilon_{A'}(\lambda) = \Pr[Y = b]\\
&\qquad\quad = \Pr[b = 0 \wedge Y = 0] + \Pr[b = 1 \wedge Y = 1]\\
&\qquad\quad = \Pr[b = 0]\Pr[Y = 0] + \Pr[b = 1]\Pr[Y = 1]\\
&\qquad\quad = \frac{1}{2}\Pr[Y = 0] + \frac{1}{2}\Pr[Y = 1]\\
&\qquad\quad = \frac{1}{2}\big(\Pr[Y = 0] + \Pr[Y = 1]\big) \leq \frac{1}{2} \cdot 1 = \frac{1}{2},\\
&\qquad |\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| \geq P^{\text{IND-CPA}}_{\Pi,D,\text{win}}(\lambda) - \frac{1}{2},\\
&\overset{(1)}{\implies} \quad \exists \text{ PPT } A \; \forall \text{ PPT } A' \; \exists \text{ PPT } \mathsf{Samp} \; \exists \text{ PT } f :\\
&\qquad |\epsilon_A - \epsilon_{A'}| \text{ non-negligible}\\
&\implies \quad \neg\text{SemanticSecure}(\Pi).
\end{aligned}$$

D : IND-CPA attacker, with its guess flipped if needed,

$P^{\text{IND-CPA}}_{\Pi,D,\text{win}}(\lambda)$: winning probability of D,

Samp : sampler choosing uniformly between D's messages,

$h = (m_0, m_1, st)$: hint independent of the hidden bit b,

f : function identifying which distinct message was sampled,

A : semantic attacker running D_1,

A' : arbitrary simulator given only $pk, h, |m_b|$.

□

By direct proof: IND-CPA $\implies$ semantic security.

$$\text{Goal:} \quad \begin{bmatrix} \forall \text{ PPT } D\, \forall \lambda \in \mathbb{N} : \\ \mathrm{Adv}^{\text{IND-CPA}}_{\Pi,D}(\lambda) \le \mathrm{negl}(\lambda) \end{bmatrix}$$

$$\implies \quad \begin{bmatrix} \forall \text{ PPT } A\, \exists \text{ PPT } A'\, \forall \text{ PPT } \mathsf{Samp}\, \forall \text{ PT } f : \\ \forall \lambda \in \mathbb{N} : \quad |\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| \le \mathrm{negl}(\lambda) \end{bmatrix}.$$

$$\text{Idea:} \quad \begin{bmatrix} \exists \text{ PPT } D\, \exists \text{ polynomial } p\, \forall \lambda \in \mathbb{N} : \\ |\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| \le p(\lambda)\, \mathrm{Adv}^{\text{IND-CPA}}_{\Pi,D}(\lambda) + \mathrm{negl}(\lambda) \end{bmatrix}$$

$$\overset{\text{IND-CPA-Secure}(\Pi)}{\implies} \quad |\epsilon_A - \epsilon_{A'}| \text{ negligible.}$$

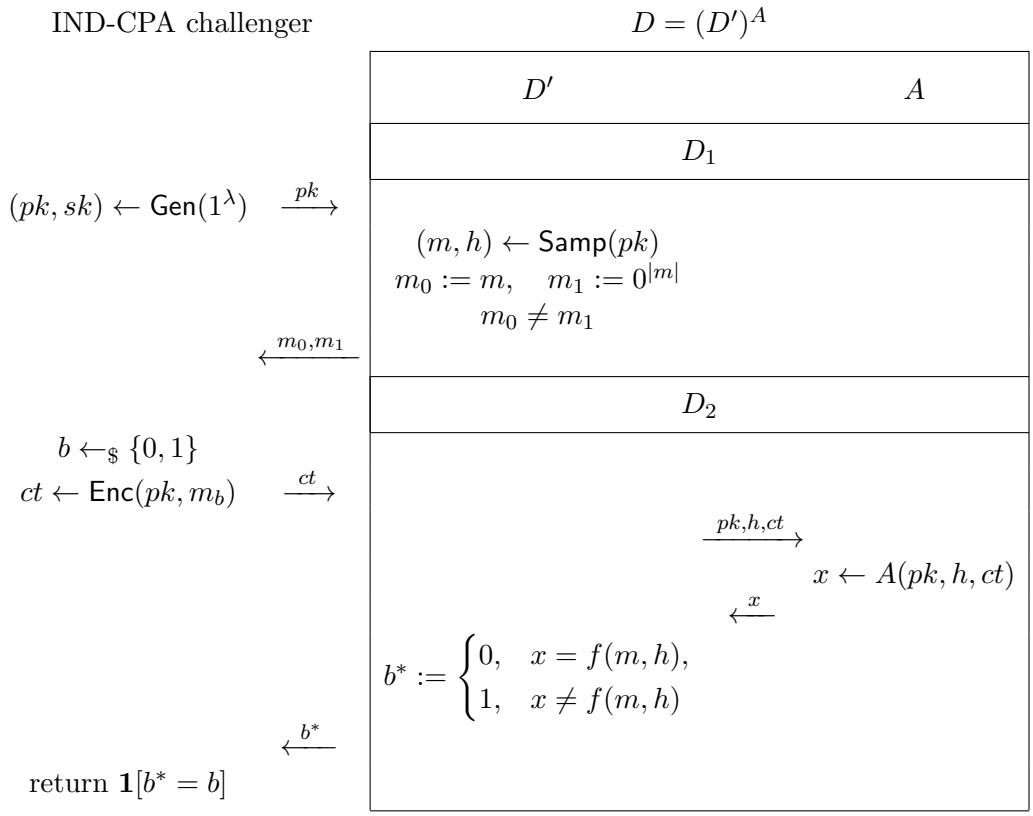
$$\forall \text{ PPT } D : \quad \underbrace{2\left|P^{\text{IND-CPA}}_{\Pi,D,\text{win}}(\lambda) - \frac{1}{2}\right|}_{\mathrm{Adv}^{\text{IND-CPA}}_{\Pi,D}(\lambda)} \overset{\text{IND-CPA-Secure}(\Pi)}{\le} \mathrm{negl}(\lambda). \tag{2}$$

Idea 2: Construct the IND-CPA game for Idea 1.

We want to construct an IND-CPA attacker D so that both ϵ_A and $\epsilon_{A'}$ appear in its winning probability. This would let us relate their difference to D's advantage, as required by Idea 1. The IND-CPA game already provides a challenge bit b with two possible values. We use these two branches to include both executions in the same game. When $b = 0$, this runs A on an encryption of m, giving the real execution. When $b = 1$, it runs A on an encryption of $0^{|m|}$, which is exactly how A' is defined below. Thus the probabilities of correctly computing $f(m, h)$ in the two branches are ϵ_A and $\epsilon_{A'}$. We can then use them to calculate D's winning probability and advantage in this one IND-CPA game, and thereby bound the semantic-security gap.

$$\forall \text{ PPT } A : \quad A'(pk, h, \ell) := A(pk, h, \mathsf{Enc}(pk, 0^\ell)).$$

$$\forall \text{ PPT } \mathsf{Samp}\, \forall \text{ PT } f : \quad \begin{array}{l} A' \text{ depends only on } A \text{ and } \Pi, \\ D = (D')^A \text{ uses the chosen } \mathsf{Samp} \text{ and } f. \end{array}$$



$$\begin{aligned}
\Pr[D \text{ wins} \mid b = 0] &= \Pr[A(pk, h, \mathsf{Enc}(pk, m)) = f(m,h)] \\
&= \epsilon_A(\lambda), \\
\Pr[D \text{ wins} \mid b = 1] &= \Pr[A'(pk, h, |m|) \neq f(m,h)] \\
&= 1 - \Pr[A'(pk, h, |m|) = f(m,h)] \\
&= 1 - \epsilon_{A'}(\lambda).
\end{aligned}$$

$$\begin{aligned}
P^{\text{IND-CPA}}_{\Pi,D,\text{win}}(\lambda) &= \Pr[D \text{ wins} \mid b = 0] \Pr[b = 0] \\
&\quad + \Pr[D \text{ wins} \mid b = 1] \Pr[b = 1] \\
&= \frac{1}{2}\epsilon_A(\lambda) + \frac{1}{2}(1 - \epsilon_{A'}(\lambda)) \\
&= \frac{1}{2} + \frac{1}{2}\big(\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)\big).
\end{aligned}$$

Direct conclusion

$$\begin{aligned}
|\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| &= 2\left|P^{\text{IND-CPA}}_{\Pi,D,\text{win}}(\lambda) - \frac{1}{2}\right| \\
&\overset{(2)}{\leq} \text{negl}(\lambda).
\end{aligned}$$

By contraposition

$$\begin{aligned}
& |\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| \text{ non-negligible} \\
\implies \quad & \text{Adv}^{\text{IND-CPA}}_{\Pi,D}(\lambda) = 2\left|P^{\text{IND-CPA}}_{\Pi,D,\text{win}}(\lambda) - \frac{1}{2}\right| \\
& \qquad = |\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| \quad \text{non-negligible} \\
& \qquad \overset{\text{IND-CPA-Secure}(\Pi)}{\implies} \perp. \\
\implies \quad & |\epsilon_A(\lambda) - \epsilon_{A'}(\lambda)| \leq \text{negl}(\lambda).
\end{aligned}$$

Questions

1. Is $|m_0| = |m_1|$ a standing assumption throughout the lectures whenever two messages are compared, or must this restriction be stated explicitly in each definition where it applies? If so, does this also apply to perfect security?
2. Would it be better to allow arbitrary PPT attackers and have the IND-CPA challenger explicitly check that $m_0, m_1 \in \mathcal{M}_\lambda$, $m_0 \neq m_1$, and $|m_0| = |m_1|$, rather than restrict to admissible attackers? If a check fails, could the game return a uniformly random bit, so the winning probability is $1/2$ and the advantage is zero?
3. Does it make sense to include explicit public key, secret key, and ciphertext spaces $\mathcal{K}_{\mathrm{pub},\lambda}$, $\mathcal{K}_{\mathrm{sec},\lambda}$, and $\mathcal{C}_\lambda$ in the PKE definition, as written above, even though the lecture specifies only the message space?
4. The lecture uses both right-pointing and left-pointing arrows, but its initial notation definition specifies the left-pointing arrow for sampling algorithm outputs. In our PKE definition, we write $\mathsf{Gen}(1^\lambda) \longrightarrow (pk, sk)$. Is this consistent with that notation definition, or should we write $(pk, sk) \leftarrow \mathsf{Gen}(1^\lambda)$ instead?
5. The exercise solutions define perfect secrecy by $\Pr[M = m \mid C = c] = \Pr[M = m]$. For PKE, should the public key also be included in the conditioning, since the attacker observes both pk and ct?

2 Lecture 2

3 Lecture 3

Definition 3.1 (IND-CCA1 game)**.**

$$A = (A_1, A_2, A_3), \qquad (st, a) := (\varepsilon, \varepsilon).$$

$$G_{\Pi,A}^{\text{IND-CCA1}}(1^\lambda)$$

Challenger		A
$(pk, sk) \leftarrow \mathsf{Gen}(1^\lambda)$	$\xrightarrow{pk}$	
Before the challenge: repeat until A_1 outputs stop		
		$(ct, st) \leftarrow A_1(pk, st, a)$
	$\xleftarrow{ct}$	$ct \neq \mathsf{stop}$
$a \leftarrow \mathsf{Dec}(sk, ct)$	$\xrightarrow{a}$	
Challenge phase: no further decryption queries		
		$(m_0, m_1, st') \leftarrow A_2(pk, st)$
		$m_0 \neq m_1 \in \mathcal{M}_\lambda, \quad \lvert m_0\rvert = \lvert m_1\rvert$
	$\xleftarrow{m_0, m_1}$	
$b \leftarrow_\$ \{0,1\}$		
$ct^* \leftarrow \mathsf{Enc}(pk, m_b)$	$\xrightarrow{ct^*}$	
	$\xleftarrow{b^*}$	$b^* \leftarrow A_3(st', ct^*)$
return $\mathbf{1}[b^* = b]$		

Π : public-key encryption scheme $(\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$,
A_1 : PPT query algorithm outputting a ciphertext (or stop) and state,
A_2 : PPT algorithm outputting m_0, m_1 and state st',
A_3 : PPT guessing stage,
st, st' : attacker states before and after message selection,
ε : empty string initializing the state and previous answer,
ct : ciphertext chosen by A for a decryption query,
a : decryption answer in $\mathcal{M}_\lambda \cup \{\perp\}$,
ct^* : challenge ciphertext,
b : uniformly random hidden bit,
b^* : bit output by A,
CCA1 : chosen-ciphertext attack with decryption access only before the challenge.

Definition 3.2 (IND-CCA1 security)**.**

$$P_{\Pi,A,\text{win}}^{\text{IND-CCA1}}(\lambda) := \Pr[G_{\Pi,A}^{\text{IND-CCA1}}(1^\lambda) = 1],$$

$$\text{Adv}_{\Pi,A}^{\text{IND-CCA1}}(\lambda) := 2\left|P_{\Pi,A,\text{win}}^{\text{IND-CCA1}}(\lambda) - \frac{1}{2}\right|,$$

$$\Pi \text{ IND-CCA1-secure} \iff \forall \text{ PPT } A\ \forall \lambda \in \mathbb{N} : \quad \text{Adv}_{\Pi,A}^{\text{IND-CCA1}}(\lambda) \leq \text{negl}(\lambda).$$

Definition 3.3 (IND-CCA1 game for Naor–Yung)**.**

$$G^{\text{IND-CCA1}}_{\Pi_{\text{NY}},A}(1^\lambda)$$

Challenger		A
$(pk_i, sk_i) \leftarrow \mathsf{Gen}(1^\lambda),\ i \in \{0,1\}$		
$crs \leftarrow \mathsf{Setup}(1^\lambda)$		
$pk := (pk_0, pk_1, crs), \quad sk := sk_0$	$\xrightarrow{pk}$	
		$(st, a) := (\varepsilon, \varepsilon)$
Before the challenge: repeat until A_1 outputs stop		
		$(ct, st) \leftarrow A_1(pk, st, a)$
	$\xleftarrow{ct=(c_0,c_1,\pi)}$	$ct \neq \mathsf{stop}$
$x := (pk_0, pk_1, c_0, c_1)$		
$a := \begin{cases} \bot, & \mathsf{Vfy}(crs, x, \pi) = 0, \\ \mathsf{Dec}(sk_0, c_0), & \mathsf{Vfy}(crs, x, \pi) = 1 \end{cases}$	$\xrightarrow{a}$	
Challenge phase: no further decryption queries		
		$(m_0, m_1, st') \leftarrow A_2(pk, st)$
		$m_0 \neq m_1 \in \mathcal{M}_\lambda$
		$\lvert m_0\rvert = \lvert m_1\rvert$
	$\xleftarrow{m_0,m_1}$	
$b \leftarrow_\$ \{0,1\}$		
$r_0^*, r_1^* \leftarrow_\$ \mathcal{R}_{\lambda,\lvert m_b\rvert}$		
$c_i^* := \mathsf{Enc}(pk_i, m_b; r_i^*),\ i \in \{0,1\}$		
$x^* := (pk_0, pk_1, c_0^*, c_1^*)$		
$w^* := (m_b, r_0^*, r_1^*)$		
$\pi^* \leftarrow \mathsf{Prove}(crs, x^*, w^*)$		
$ct^* := (c_0^*, c_1^*, \pi^*)$	$\xrightarrow{ct^*}$	
	$\xleftarrow{b^*}$	$b^* \leftarrow A_3(st', ct^*)$
return $\mathbf{1}[b^* = b]$		

Π_{NY} : Naor–Yung encryption scheme built from Π,
$\mathsf{Setup}, \mathsf{Prove}, \mathsf{Vfy}$: NIZK reference-string, proving, and verification algorithms,
$\mathcal{R}_{\lambda,\ell}$: encryption randomness space for messages of length ℓ,
x^* : statement that c_0^*, c_1^* encrypt the same message,
w^* : witness containing that message and both encryption coins.

Malformed decryption queries return $\bot$.

Theorem 3.4 (IND-CCA1 security of Naor–Yung)**.**

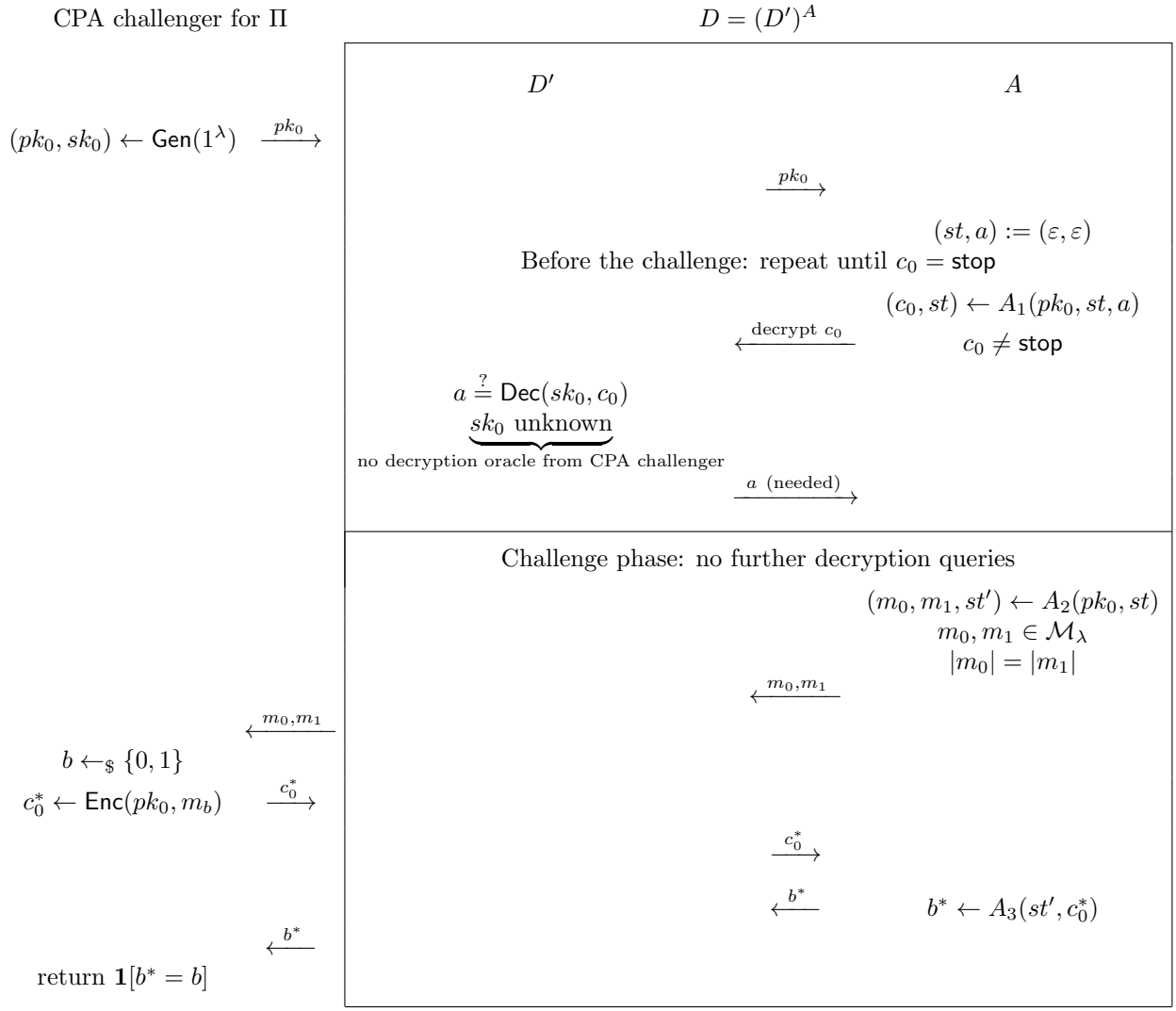
$$\Pi \textit{ IND-CPA-secure}, \quad \mathsf{NIZK} \textit{ complete, sound, and zero-knowledge for } R_{\mathrm{NY}} \implies \Pi_{\mathrm{NY}} \textit{ IND-CCA1-secure.}$$

$$R_{\mathrm{NY}} := \big\{ \big((pk_0, pk_1, c_0, c_1), (m, r_0, r_1) \big) : c_0 = \mathsf{Enc}(pk_0, m; r_0) \ \wedge \ c_1 = \mathsf{Enc}(pk_1, m; r_1) \big\},$$

Π_{NY} : *Naor–Yung construction using* Π *and* NIZK.

Naor–Yung: answering decryption queries in the reduction

Desired reduction with only c_0: the decryption step is missing.

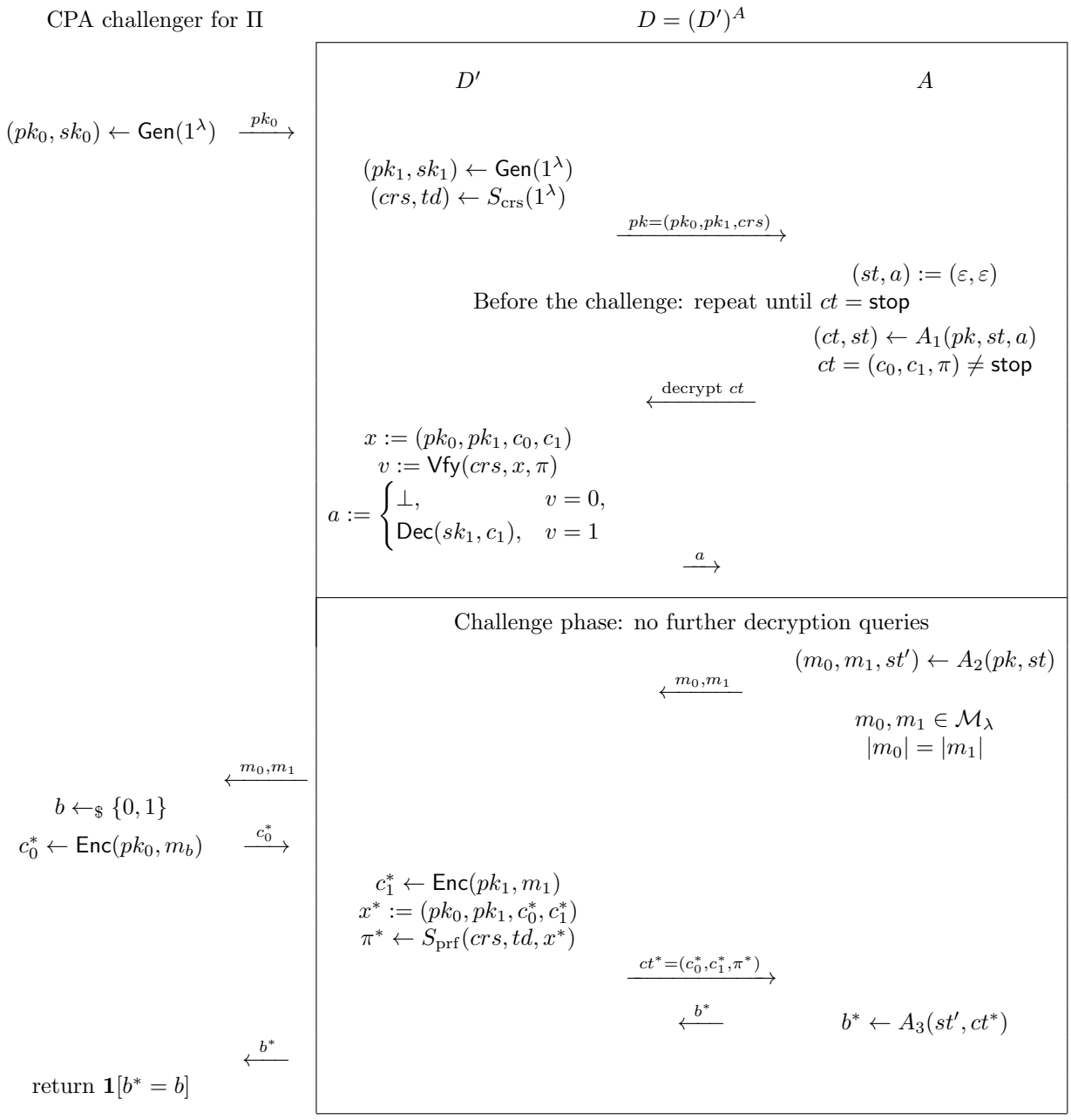


c_0 : ciphertext chosen by A for a decryption query,
a : decryption answer, a plaintext or $\perp$,
c_0^* : challenge ciphertext,
b : hidden challenge bit,
b^* : bit output by A.

The question mark is the missing step: how can D' answer the decryption query without sk_0? The challenge forwarding is already specified, but this is not a working reduction until the decryption queries can be answered.

CCA1 attack A on Π_{NY} $\xrightarrow{\text{reduction}}$ CPA attack D on Π.

Complete CPA interaction for the hop $G_3 \to G_4$:



A : CCA1 attacker choosing ciphertexts to query,
D : CPA reduction running A,
D' : wrapper answering queries and constructing the challenge for A,
sk_0 : secret key unknown to D,
sk_1 : independent secret key generated and known by D,
π : proof that c_0, c_1 encrypt the same message,
$S_{\mathrm{crs}}, S_{\mathrm{prf}}$: NIZK simulators for the reference string and proof,
td : simulation trapdoor known to D,
π^* : simulated proof attached to the challenge,
$\perp$: rejected decryption query.

$$c_0 = \mathsf{Enc}(pk_0, m; r_0), \quad c_1 = \mathsf{Enc}(pk_1, m; r_1) \quad \Longrightarrow \quad \underbrace{\mathsf{Dec}(sk_0, c_0)}_{\text{actual scheme}} = m = \underbrace{\mathsf{Dec}(sk_1, c_1)}_{\text{reduction } D}.$$

$$b = 0 \implies A \text{ sees } G_3, \qquad b = 1 \implies A \text{ sees } G_4.$$

This is one hop of the proof. The simulated proof lets D construct the challenge without knowing m_b or the randomness used for c_0^*; in G_3, the two challenge components may encrypt different messages. The separate soundness hops justify switching the decryption key for queries made before the challenge.

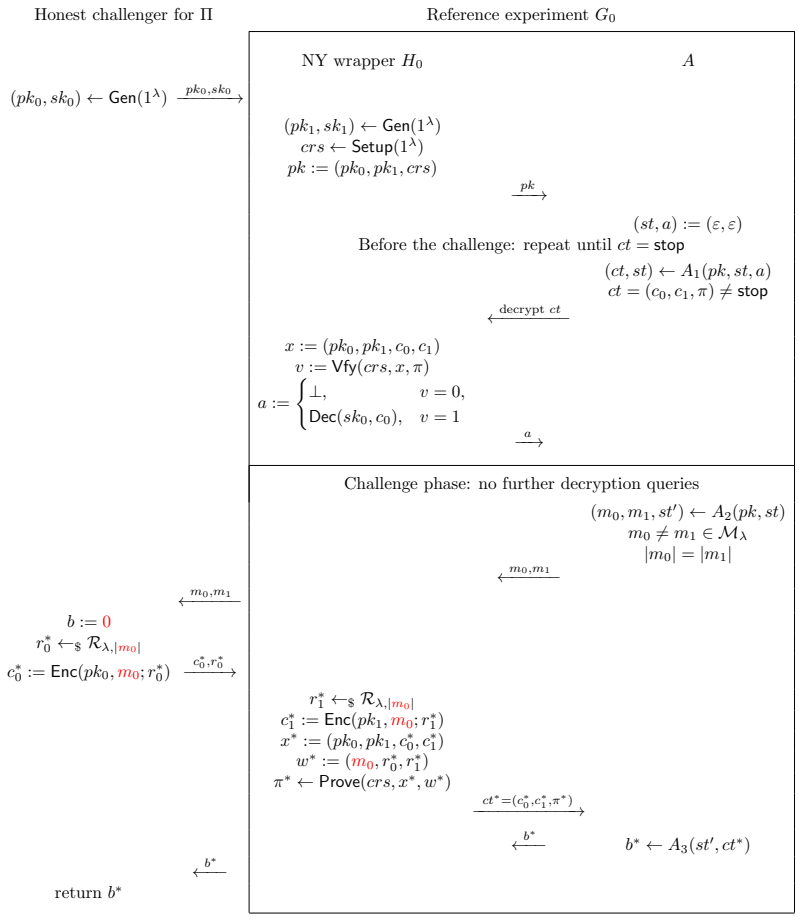
Naor–Yung: sequence of games

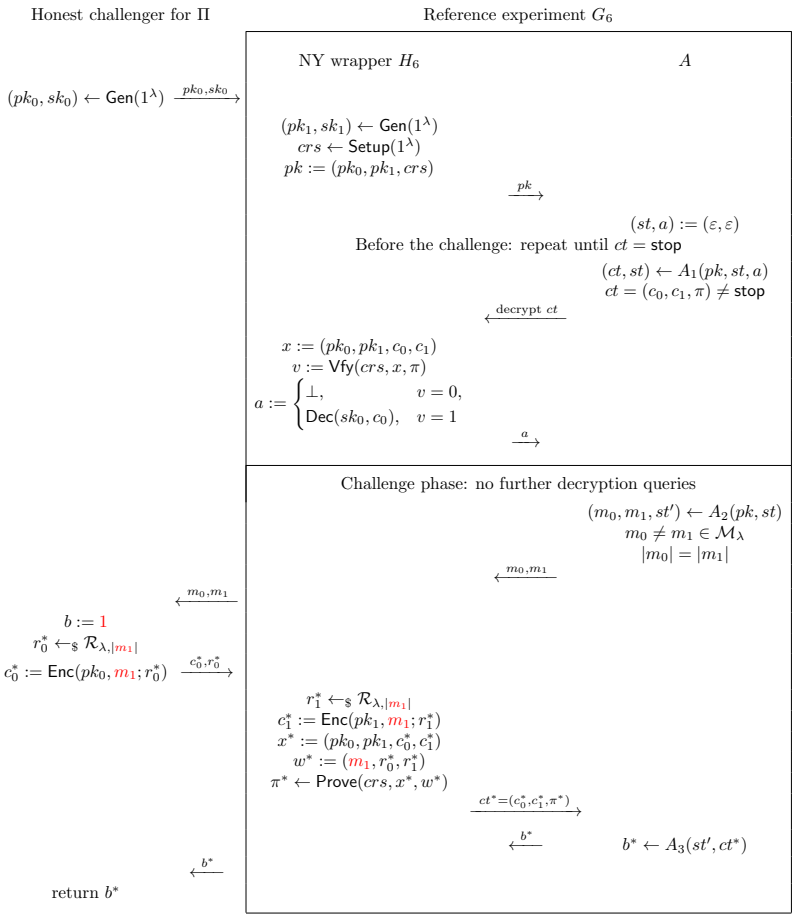
Goal: $|\Pr[G_0 = 1] - \Pr[G_6 = 1]| \leq \mathrm{negl}(\lambda)$.

Game	c_0^* encrypts	c_1^* encrypts	crs, π^*	decryption key	Change from previous row
G_0	m_0	m_0	real	sk_0	initial game
G_1	m_0	m_0	simulated	sk_0	simulate crs, π^* (ZK)
G_2	m_0	m_1	simulated	sk_0	change c_1^* (CPA under pk_1)
G_3	m_0	m_1	simulated	sk_1	switch to sk_1 (soundness)
G_4	m_1	m_1	simulated	sk_1	change c_0^* (CPA under pk_0)
G_5	m_1	m_1	simulated	sk_0	switch to sk_0 (soundness)
G_6	m_1	m_1	real	sk_0	restore real crs, π^* (ZK)

G_i : experiment specified by row i, outputting A's final bit,
G_0 : actual Naor–Yung CCA1 interaction with challenge bit fixed to 0,
G_6 : actual Naor–Yung CCA1 interaction with challenge bit fixed to 1,
decryption key : key used to answer pre-challenge queries after verification,
ZK : zero knowledge of the NIZK proof system,
red : entry changed from the previous row.

Definition 3.5 (Endpoint games G_0 and G_6).





G_0, G_6 : real Naor–Yung experiments with challenge bit fixed to 0 and 1,

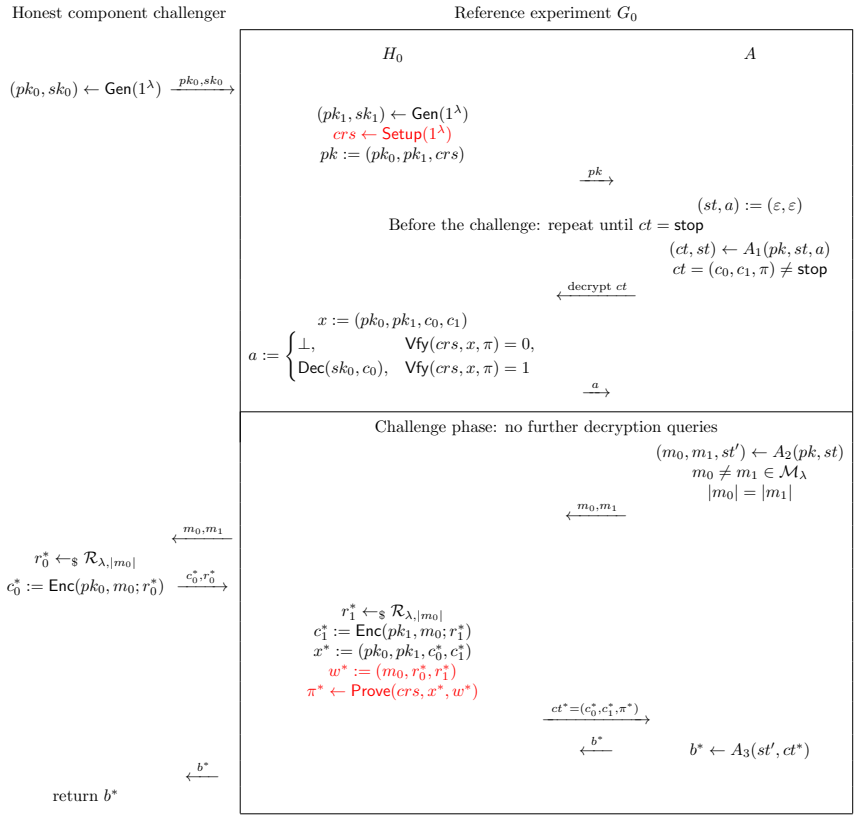
H_0, H_6 : NY wrappers receiving sk_0, r_0^* to decrypt and generate real proofs,

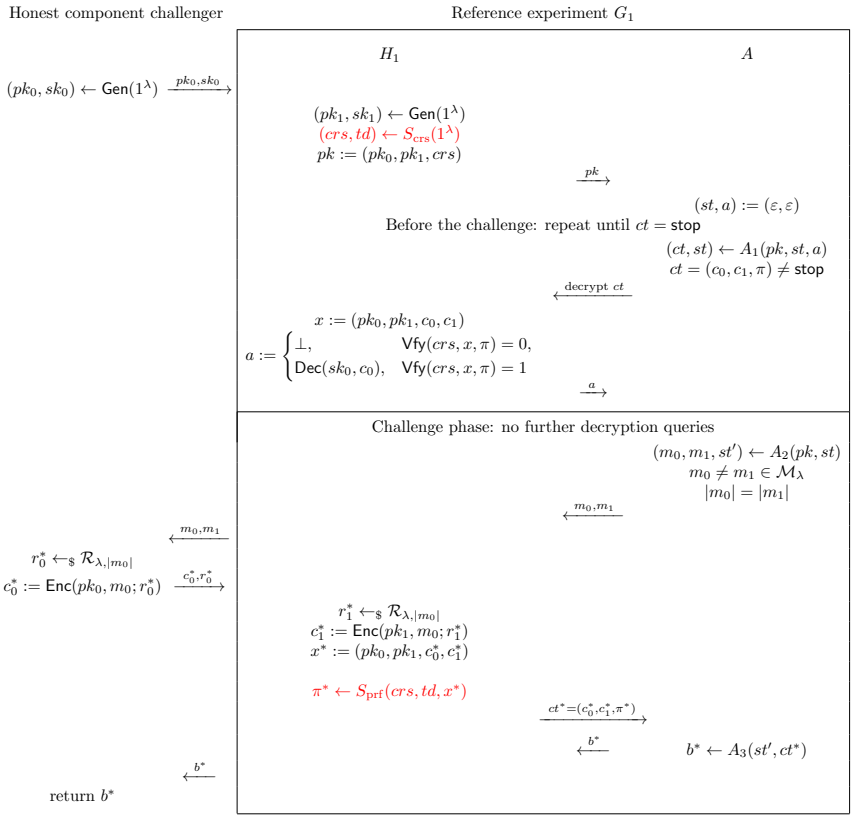
red : entries that differ between the two experiments,

b^* : final bit output by A and returned by each experiment.

The honest challenger shares sk_0 and the challenge coins r_0^* with the wrapper; A receives neither. Malformed queries return $\bot$.

Comparison $G_0 \to G_1$



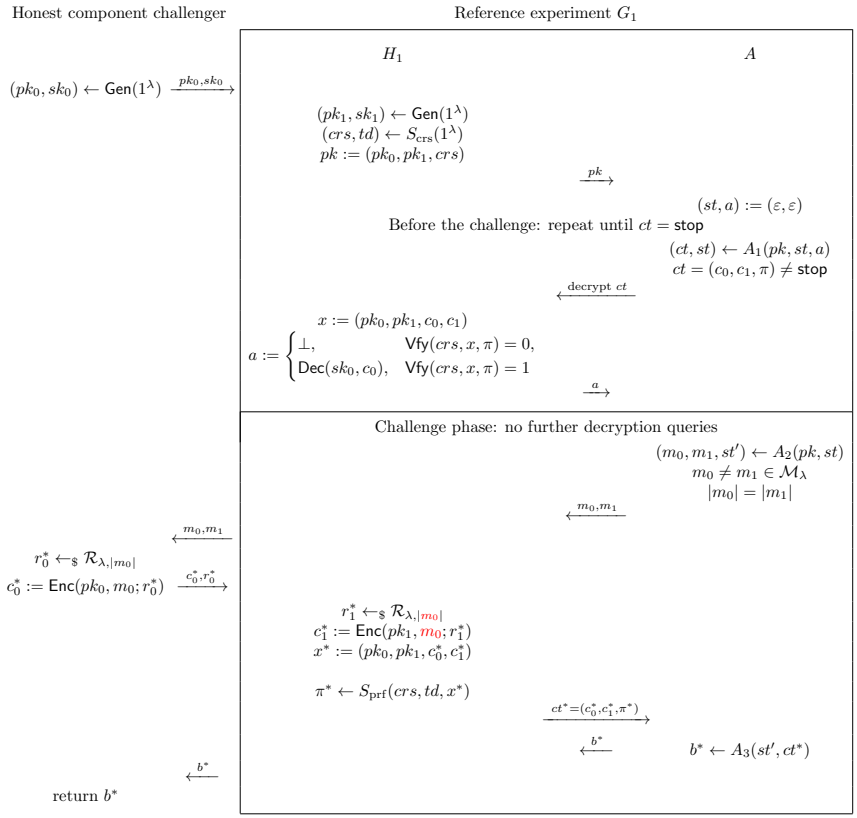


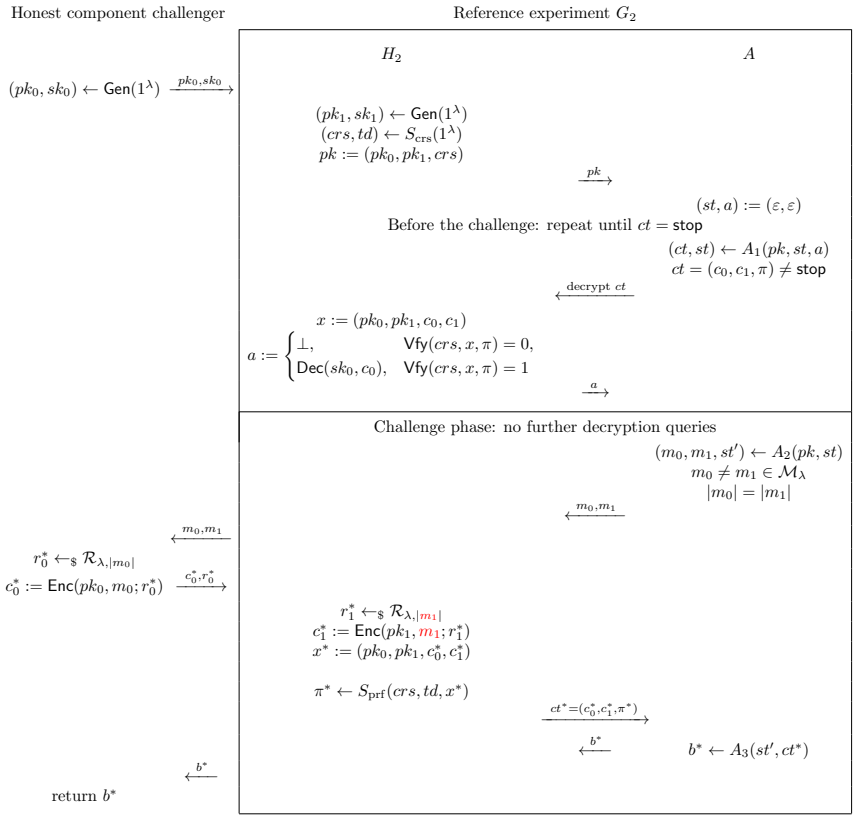
$$q_k(\lambda) := \Pr[G_k(1^\lambda) = 1], \qquad \epsilon_1(\lambda) := |q_1(\lambda) - q_0(\lambda)|.$$

$$\begin{aligned} \epsilon_1(\lambda) &= \left|\Pr[\mathsf{Exp}^{\mathrm{real}}_{Z_{01}} = 1] - \Pr[\mathsf{Exp}^{\mathrm{ideal}}_{Z_{01}} = 1]\right| \\ &= \mathrm{Adv}^{\mathrm{ZK}}_{\mathsf{NIZK}, Z_{01}}(\lambda) \leq \mathrm{negl}(\lambda). \end{aligned}$$

Z_{01} runs A, generates both key pairs, and uses its ZK challenger for crs, π^*. The challenge statement is true: both components encrypt m_0. Its output is A's bit.

Comparison $G_1 \to G_2$





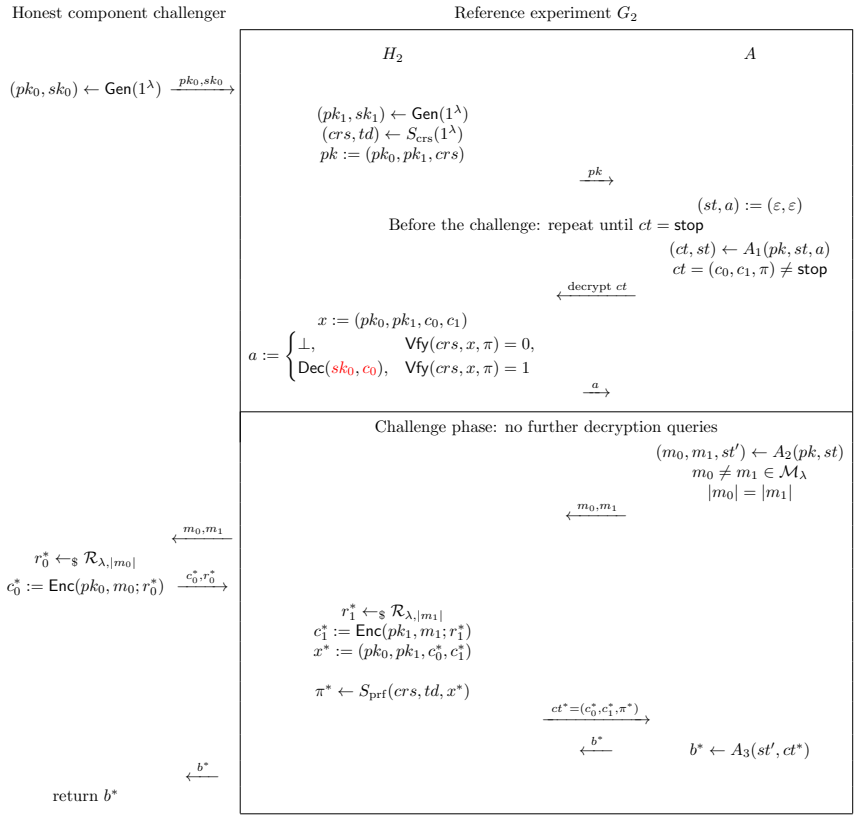
$$q_k(\lambda) := \Pr[G_k(1^\lambda) = 1], \qquad \epsilon_2(\lambda) := |q_2(\lambda) - q_1(\lambda)|.$$

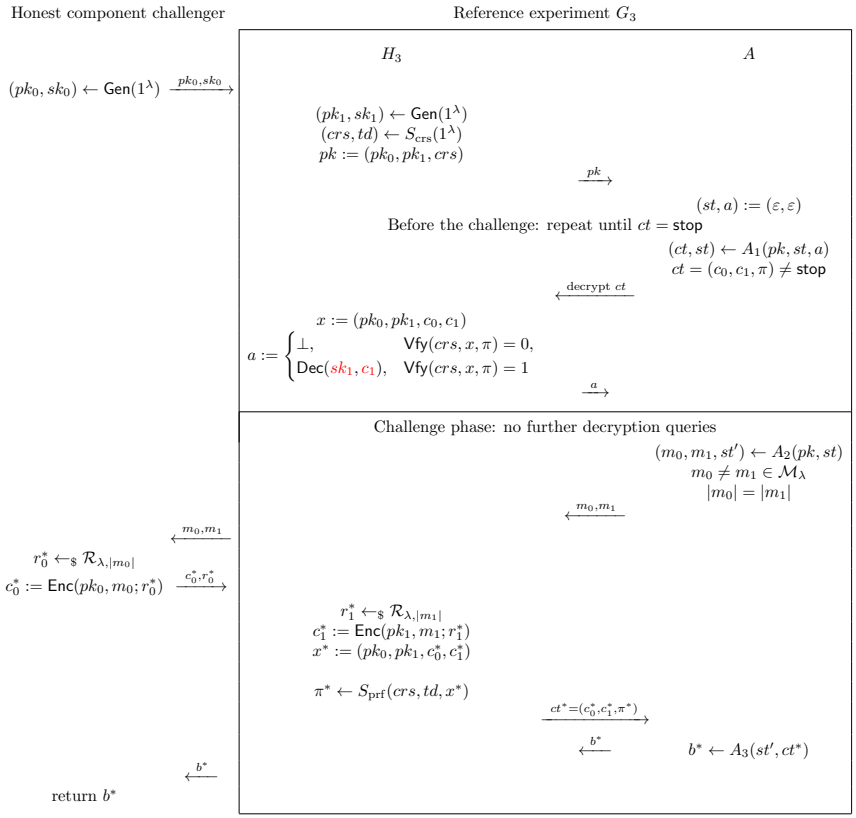
$$\Pr[C_{12} \text{ wins}] = \frac{1}{2}(1 - q_1) + \frac{1}{2}q_2 = \frac{1}{2} + \frac{1}{2}(q_2 - q_1),$$

$$\mathrm{Adv}^{\text{IND-CPA}}_{\Pi, C_{12}}(\lambda) = 2\left|\Pr[C_{12} \text{ wins}] - \frac{1}{2}\right| = |q_2 - q_1| = \epsilon_2(\lambda) \leq \mathrm{negl}(\lambda).$$

C_{12} embeds its CPA challenge under pk_1 and knows sk_0 for decryption queries. Its hidden CPA bit $\beta = 0$ gives G_1; $\beta = 1$ gives G_2. It forwards A's output as its guess.

Comparison $G_2 \to G_3$





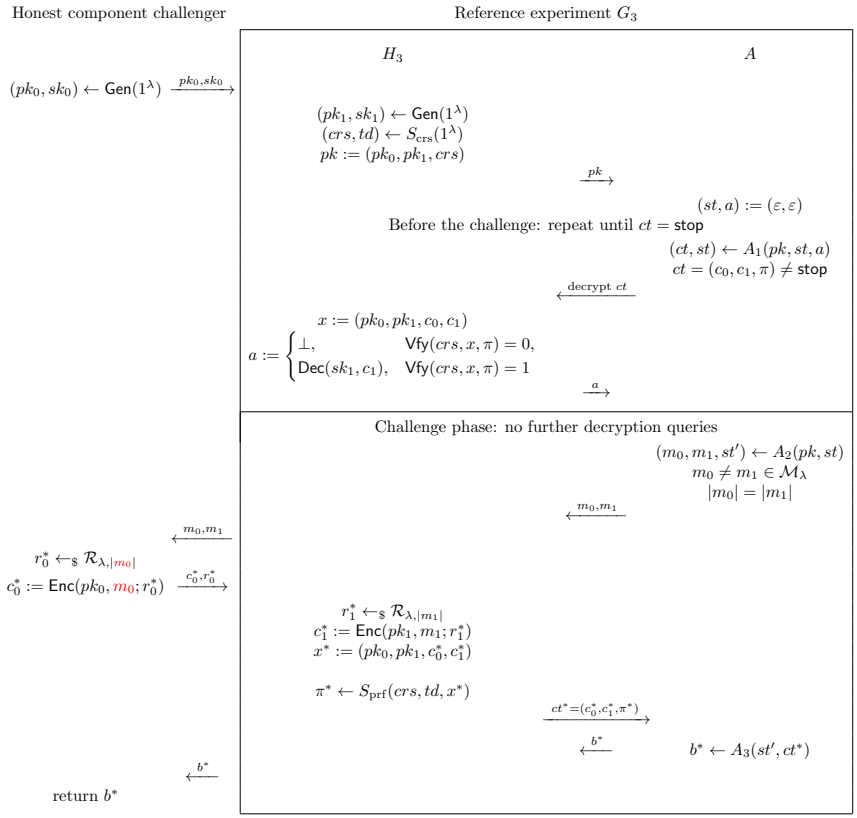
$$q_k(\lambda) := \Pr[G_k(1^\lambda) = 1], \qquad \epsilon_3(\lambda) := |q_3(\lambda) - q_2(\lambda)|.$$

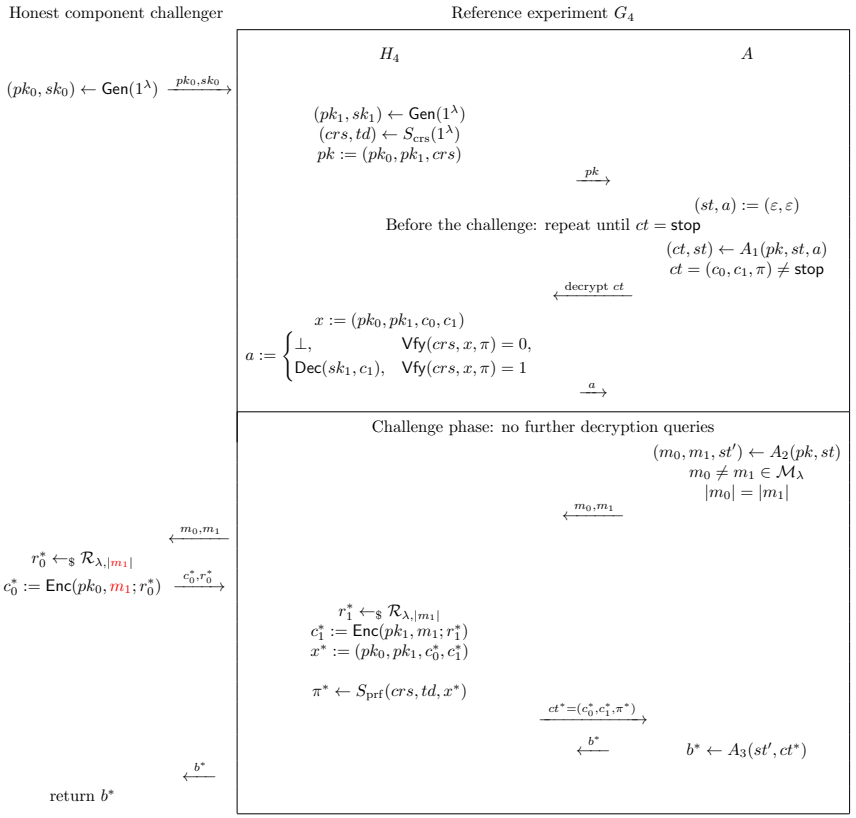
$$\mathsf{Bad} := \{\exists \text{ pre-challenge query } (c_0, c_1, \pi) : \ \mathsf{Vfy}(crs, x, \pi) = 1 \ \wedge \ \mathsf{Dec}(sk_0, c_0) \neq \mathsf{Dec}(sk_1, c_1)\},$$

$$\begin{aligned} \epsilon_3(\lambda) &\leq \Pr[\mathsf{Bad} \text{ in } G_2] \\ &\leq \mathrm{Adv}^{\mathrm{sound}}_{\mathsf{NIZK}, B_{23}}(\lambda) + \mathrm{Adv}^{\mathrm{ZK}}_{\mathsf{NIZK}, Z_{23}}(\lambda) \leq \mathrm{negl}(\lambda). \end{aligned}$$

Until Bad, the two decryption answers agree (using perfect correctness). B_{23} outputs an accepted inconsistent query under an honest crs; Z_{23} bounds the change in $\Pr[\mathsf{Bad}]$ when replacing the simulated crs by an honest one. All these queries precede the simulated challenge proof.

Comparison $G_3 \to G_4$





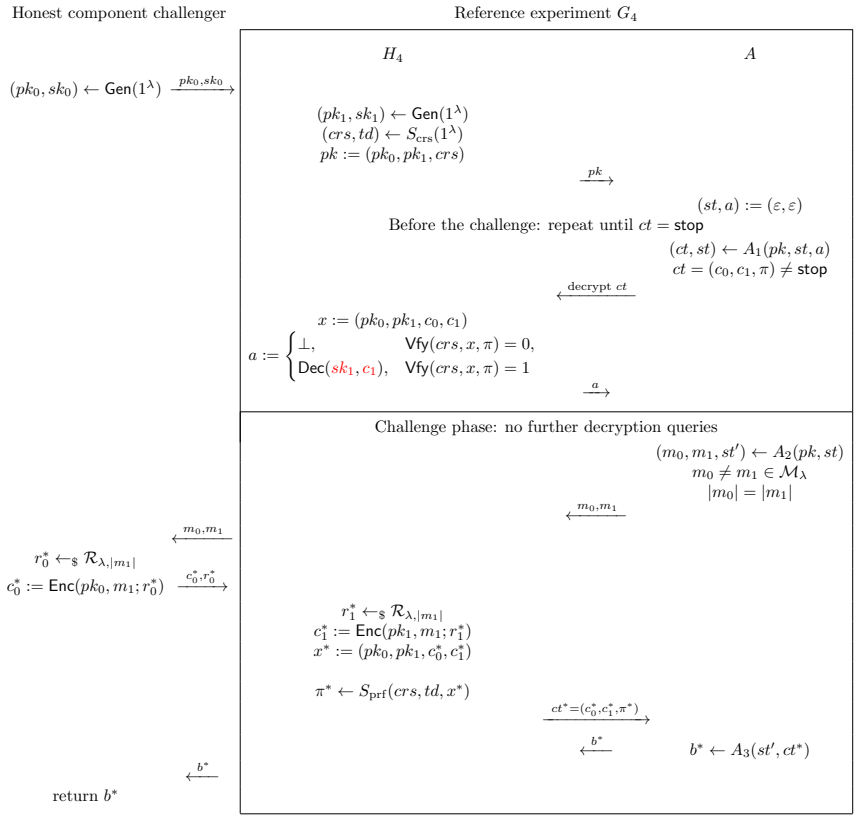
$$q_k(\lambda) := \Pr[G_k(1^\lambda) = 1], \qquad \epsilon_4(\lambda) := |q_4(\lambda) - q_3(\lambda)|.$$

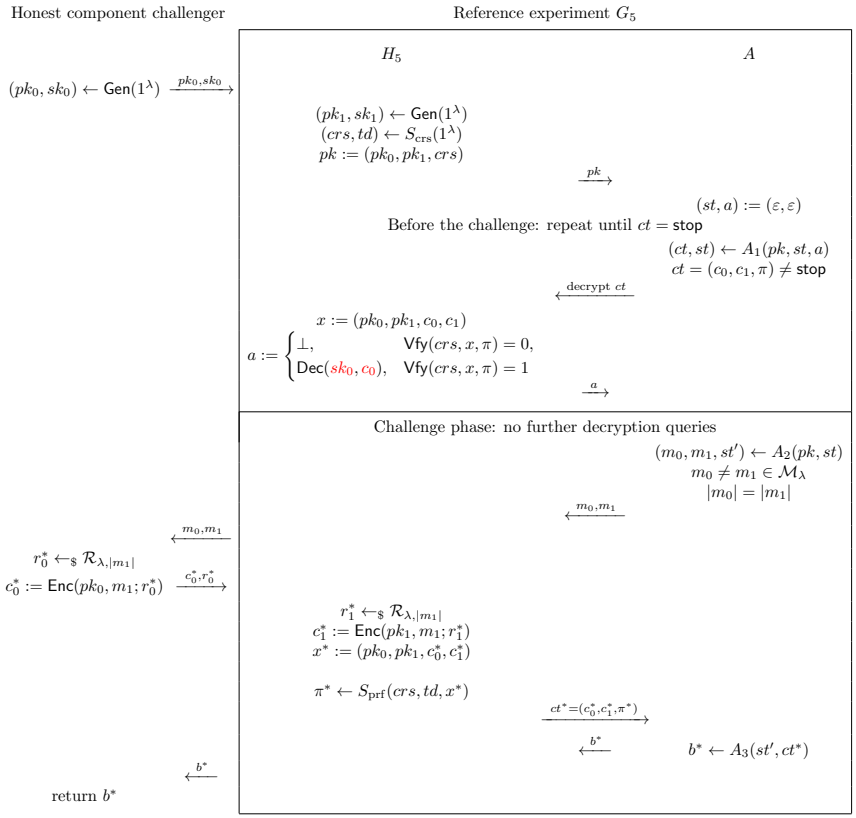
$$\Pr[C_{34} \text{ wins}] = \frac{1}{2}(1 - q_3) + \frac{1}{2}q_4 = \frac{1}{2} + \frac{1}{2}(q_4 - q_3),$$

$$\mathrm{Adv}^{\text{IND-CPA}}_{\Pi, C_{34}}(\lambda) = 2\left|\Pr[C_{34} \text{ wins}] - \frac{1}{2}\right| = |q_4 - q_3| = \epsilon_4(\lambda) \leq \mathrm{negl}(\lambda).$$

C_{34} embeds its CPA challenge under pk_0 and knows sk_1 for decryption queries. Its hidden CPA bit $\beta = 0$ gives G_3; $\beta = 1$ gives G_4. It forwards A's output as its guess.

Comparison $G_4 \to G_5$





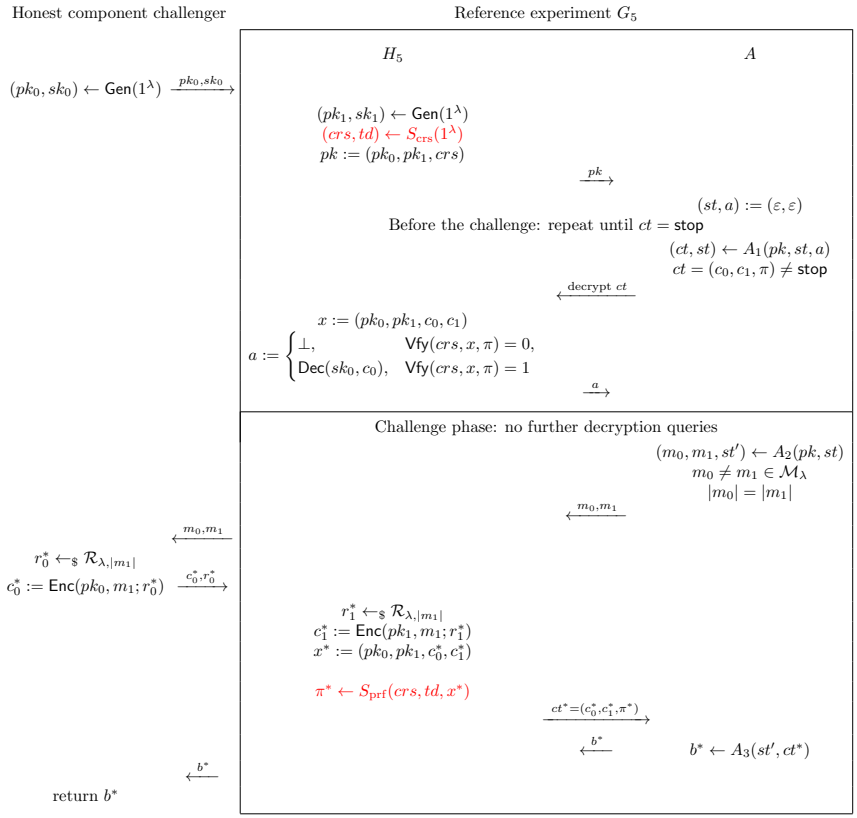
$$q_k(\lambda) := \Pr[G_k(1^\lambda) = 1], \qquad \epsilon_5(\lambda) := |q_5(\lambda) - q_4(\lambda)|.$$

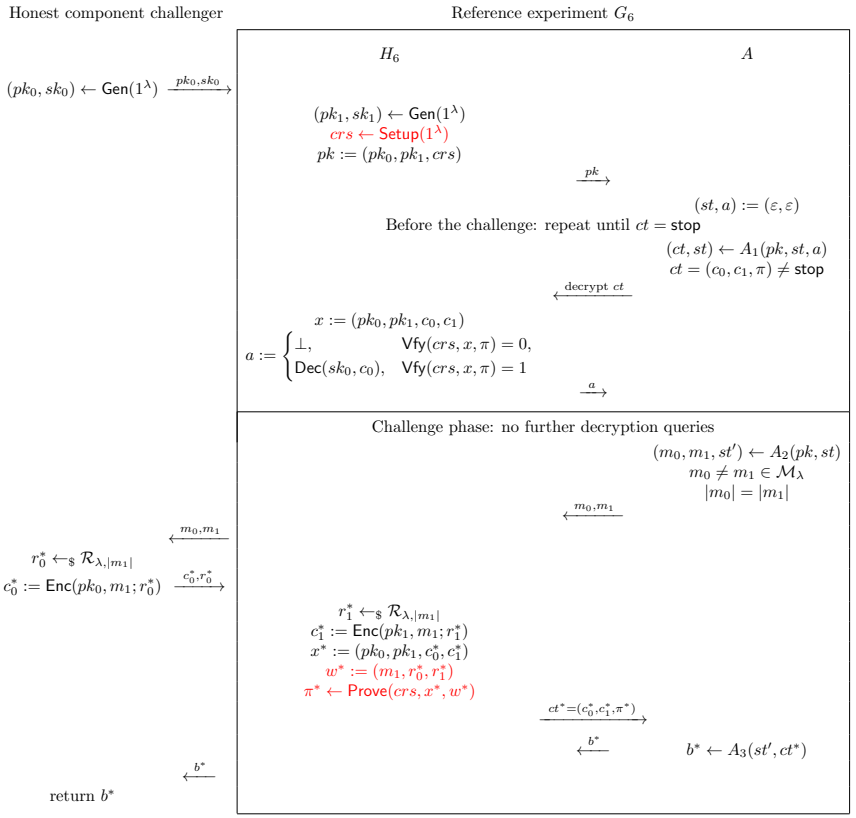
$$\mathsf{Bad} := \{\exists \text{ pre-challenge query } (c_0, c_1, \pi) : \ \mathsf{Vfy}(crs, x, \pi) = 1 \ \wedge \ \mathsf{Dec}(sk_0, c_0) \neq \mathsf{Dec}(sk_1, c_1)\},$$

$$\begin{aligned} \epsilon_5(\lambda) &\leq \Pr[\mathsf{Bad} \text{ in } G_4] \\ &\leq \mathrm{Adv}^{\mathrm{sound}}_{\mathsf{NIZK}, B_{45}}(\lambda) + \mathrm{Adv}^{\mathrm{ZK}}_{\mathsf{NIZK}, Z_{45}}(\lambda) \leq \mathrm{negl}(\lambda). \end{aligned}$$

Until Bad, the two decryption answers agree (using perfect correctness). B_{45} outputs an accepted inconsistent query under an honest crs; Z_{45} bounds the change in $\Pr[\mathsf{Bad}]$ when replacing the simulated crs by an honest one. All these queries precede the simulated challenge proof.

Comparison $G_5 \to G_6$





$$q_k(\lambda) := \Pr[G_k(1^\lambda) = 1], \qquad \epsilon_6(\lambda) := |q_6(\lambda) - q_5(\lambda)|.$$

$$\begin{aligned} \epsilon_6(\lambda) &= \left|\Pr[\mathsf{Exp}^{\mathrm{real}}_{Z_{56}} = 1] - \Pr[\mathsf{Exp}^{\mathrm{ideal}}_{Z_{56}} = 1]\right| \\ &= \mathrm{Adv}^{\mathrm{ZK}}_{\mathsf{NIZK}, Z_{56}}(\lambda) \leq \mathrm{negl}(\lambda). \end{aligned}$$

Z_{56} runs A, generates both key pairs, and uses its ZK challenger for crs, π^*. The challenge statement is true: both components encrypt m_1. Its output is A's bit.

Combining the six bounds

$$\begin{aligned}
\mathrm{Adv}^{\text{IND-CCA1}}_{\Pi_{\mathrm{NY}},A}(\lambda) &= |q_6(\lambda) - q_0(\lambda)| \\
&\leq \sum_{i=1}^{6} \epsilon_i(\lambda) \\
&\leq \mathrm{Adv}^{\text{IND-CPA}}_{\Pi,C_{12}}(\lambda) + \mathrm{Adv}^{\text{IND-CPA}}_{\Pi,C_{34}}(\lambda) \\
&\quad + \mathrm{Adv}^{\text{sound}}_{\mathsf{NIZK},B_{23}}(\lambda) + \mathrm{Adv}^{\text{sound}}_{\mathsf{NIZK},B_{45}}(\lambda) \\
&\quad + \mathrm{Adv}^{\text{ZK}}_{\mathsf{NIZK},Z_{01}}(\lambda) + \mathrm{Adv}^{\text{ZK}}_{\mathsf{NIZK},Z_{23}}(\lambda) \\
&\quad + \mathrm{Adv}^{\text{ZK}}_{\mathsf{NIZK},Z_{45}}(\lambda) + \mathrm{Adv}^{\text{ZK}}_{\mathsf{NIZK},Z_{56}}(\lambda) \\
&\leq \mathrm{negl}(\lambda).
\end{aligned}$$

$q_i(\lambda)$: probability that A outputs 1 in G_i,

$\epsilon_i(\lambda)$: absolute difference between the output probabilities of G_{i-1}, G_i,

$\mathrm{Adv}^{\mathrm{ZK}}$: absolute difference of real and ideal ZK output probabilities,

$\mathrm{Adv}^{\mathrm{sound}}$: probability of producing an accepted false statement under an honest crs,

C_{12}, C_{34} : PPT reductions to CPA security using A,

B_{23}, B_{45} : PPT reductions to soundness using A,

$Z_{01}, Z_{23}, Z_{45}, Z_{56}$: PPT reductions to zero knowledge using A.